

If...then & Biconditionals

Name: _____ Date: _____

Patience rule: When working with conditionals, write the contrapositive explicitly — do not swap parts from memory. For biconditionals, check *both* directions separately before declaring T or F.

Deduction problems — Conditionals

D4.1 DEDUCTION

Original conditional: “If a number is divisible by 12, then it is divisible by 4.”

1. Define p and q in symbols.
2. Write the **converse**, **inverse**, and **contrapositive** in English and in symbols.
3. For each of the four forms (original + three relatives), give a concrete integer example that shows whether *that form’s* implication holds or fails. You need four different examples — one per form — with arithmetic shown.
4. Build a 4-row truth table for the original $p \rightarrow q$ and a separate 4-row table for the contrapositive. Confirm they match row by row.

FOUR FORMS + FOUR NUMERIC EXAMPLES + TWO TABLES

D4.2 DEDUCTION

Three chained claims about a single unknown truth value assignment to A , B , C (treat each letter as “event A happens,” etc.):

- C1: If A then B .
 - C2: If B then C .
 - C3: If C then not A .
1. Assume A is **true**. Follow the chain step by step (write each deduction). What must be true about B and C ? Do you reach a contradiction?
 2. Therefore, what must A be? Justify.
 3. With A fixed, list **all** assignments to (B, C) that make C1, C2, and C3 all true simultaneously. Use an exhaustive 4-row table for (B, C) if helpful.
 4. Translate C1–C3 into one English story (three sentences max) that matches your solution.

CHAIN REASONING + COMPLETE (B,C) LIST + STORY

D4.3 DEDUCTION

Build a full 8-row truth table for:

$$(p \leftrightarrow q) \rightarrow (r \rightarrow p)$$

Include columns for $p \leftrightarrow q$, $r \rightarrow p$, and the final conditional.

1. How many rows make the final column false? List them.
2. For each false row, explain in English why the outer conditional failed (remember: false only when hypothesis T and conclusion F).
3. Does any row have $p \leftrightarrow q$ true while $r \rightarrow p$ is false? If yes, identify all such rows.

8-ROW TABLE + FALSE-ROW ANALYSIS

D4.4 DEDUCTION

A librarian gives you five clues about a hidden book (genre is mystery or sci-fi; length is short or long; cover is red or blue). Each clue is a conditional or biconditional:

1. If the book is sci-fi, then the cover is blue.
2. If the cover is red, then the book is long.
3. The book is short if and only if it is mystery.
4. If the book is long and mystery, then the cover is not blue.
5. If the book is sci-fi, then it is not long.

Define letters for the four atomic facts (mystery, sci-fi, short, long, red, blue — use as many letters as you need, but be consistent).

- a. Translate all five clues into symbols.
- b. List every consistent assignment of genre, length, and cover that satisfies all five. Check each candidate against every clue — no skipping.
- c. Is the hidden book uniquely determined? Prove your answer.

SYMBOLIZATION + CASE CHECK + UNIQUENESS PROOF